



Sustainable Income Optimization Model Through Food and Livestock Commodity Trade Integration in the Indonesia–Timor Leste Border Area

Boanerges Putra Sipayung^{1*}, Umbu Joka¹, Achmad Subchiandi Maulana¹,
Agustinus Nubatonis¹, Yohanes Pv Mambur¹, Fried Markus Allung Blegur²,
Rifqi Adi Darmawan³, Piki Darma Kristian Pardede⁴

¹ Agribusiness Program, Faculty of Agricultural, Science, and Health, Timor University, 85614 Kefamenanu, Indonesia

² Mathematic Program, Faculty of Agricultural, Science, and Health, Timor University, 85614 Kefamenanu, Indonesia

³ Ministry of Agrarian Affairs and Spatial Planning National Land Agency, 85614 Kefamenanu, Indonesia

⁴ Public Administration Program, Faculty of Social and Political Science, Timor University, 85614 Kefamenanu, Indonesia

* Correspondence: Boanerges Putra Sipayung (sipayung.boanerges@gmail.com)

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Abstract: Increased agricultural production could improve household income but often generates adverse environmental impacts, including soil degradation, rising temperatures, and drought, thereby contributing to climate change. This study aims to optimize income and carbon emissions in the trade of rice, corn, and cattle commodities in the Indonesia–Timor Leste border region and to assess the performance of integrated sustainable trade among farmers, traders, and processing industries. A Multi-Objective Linear Programming (MOLP) model and Partial Least Squares Structural Equation Modeling (PLS-SEM) were employed for analysis. The findings indicated that increased trade activities could improve economic outcomes while maintaining emissions within manageable limits. Farmer income is projected to increase by IDR 5.779 billion per production season, with improved cost efficiency at approximately IDR 64,000 per acre and maximum emissions of 356,561 tons CO₂e. Traders' income is expected to increase by IDR 8.526 billion, with maximum emissions of 2,443.241 tons CO₂e and average transport costs of IDR 4,600 per kilometer. Carbon emissions at the farm level primarily stem from inefficient use of fertilizer and land burning, while emissions at the trader level are driven by transport capacity and travel distance. Although processing industries have established direct relationships with farmers, most farmers remain dependent on traders for market access. Strengthening the capacity of processing industries in the border region is therefore considered essential for maximizing farmers' income.

Keywords: Agricultural-livestock trade; Border region; Carbon emissions; Commodities; Sustainability

1. Introduction

Infrastructural development in the Indonesia–Timor Leste border region is intended to enhance public accessibility and stimulate economic growth. Empirical evidence showed that infrastructure investment contributed to long-term economic development and reduced income inequality (Nugraha et al., 2020; Taena et al., 2023b; Zhang & Cheng, 2023). The local economy in this region is predominantly supported by traditional food crop farming and cattle breeding, characterized by low levels of education, productivity, and a high poverty rate among farmers (Sipayung et al., 2023; Taena et al., 2023a). The border region functions as a key production center for rice, corn, and beef cattle. Production of rice increased from 131,000 tons in 2020 to 164,000 tons in 2022,

while corn production rose from 167,000 tons to 193,000 tons over the same period. Similarly, the cattle population increased from 575,000 heads in 2020 to 631,000 heads in 2022 (Badan Pusat Statistik, 2023).

Development of infrastructure and increased production directly increase trade flows, which impact parties involved in the supply chain. Although rising agricultural output and infrastructure contribute to higher household income, it also generates adverse environmental impacts, including soil degradation, drought, and increased carbon emissions (Waithaka, 2023). These environmental effects reflect the inverse relationship between production growth and sustainability (Agovino et al., 2019; Raihan, 2023), especially when production is driven by traditional farming practices and limited environmental awareness (Waseem et al., 2020). Increased production also tends to intensify trade activities, further elevating carbon emissions (Chen et al., 2021). Sustainable development should address social, economic, and ecological aspects in a balanced manner (Fauzi, 2019; Purvis et al., 2019).

This study aims to optimize both income and carbon emission levels associated with the trade of rice, corn, and cattle in the Indonesia–Timor Leste border region, and to evaluate the sustainability performance of these commodity trade systems.

2. Methodology

2.1 Research Site and Setting

This study was conducted in Indonesia's border region with Timor Leste which can be seen in Figure 1, encompassing four regencies: Kupang, North Central Timor, Belu, and Malaka. The research was carried out from July 2024 to December 2024.

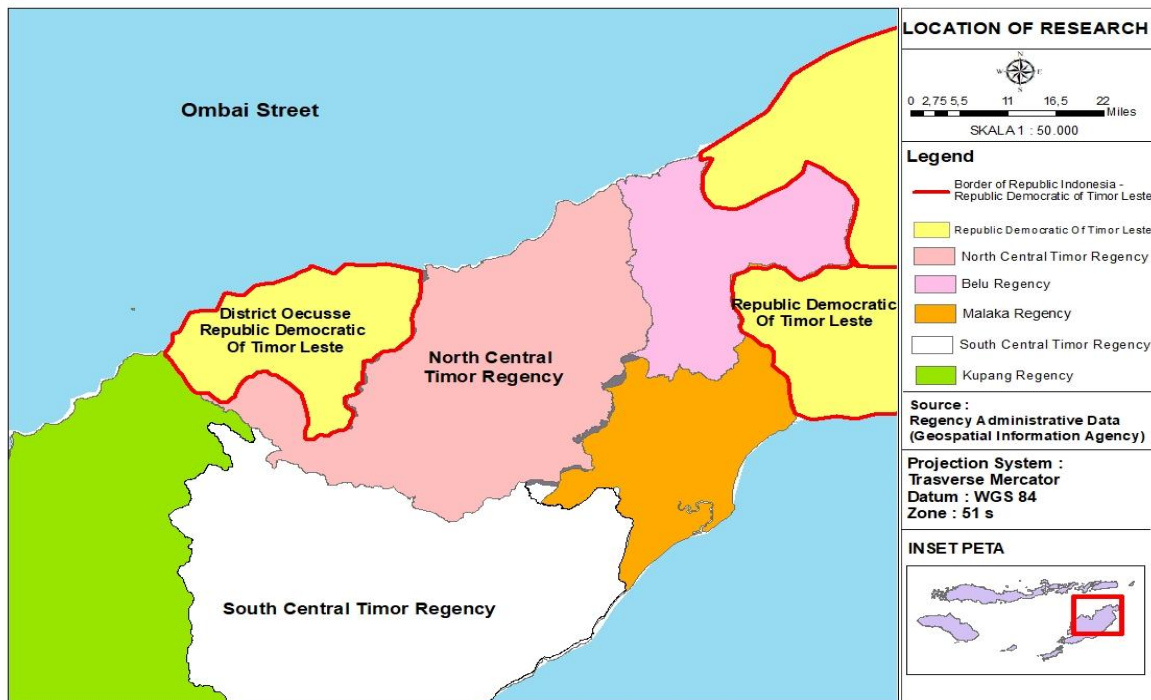


Figure 1. Location of the research

2.2 Data Collection

This study examined both primary and secondary data. Primary data were collected through interviews with key marketing institutions in the border area, including food crop farmers who also raise cattle, food traders, livestock traders, processing industry actors, and consumers. The total sample consisted of 500 respondents: 100 food crop farmers with cattle, 100 food traders, 100 livestock traders, 100 processing industry representatives, and 100 consumers. Secondary data were obtained from the East Nusa Tenggara Provincial Agriculture Office, the Central Statistics Agency, and the East Nusa Tenggara Provincial Directorate of Land Transportation.

2.3 Data Analysis

The supply chain system consists of three production nodes (rice, corn, and livestock farmers), intermediate

distribution warehouses, and consumers. Product flow can be direct from farmers to consumers or indirect through warehouses, depending on production capacity, storage, transportation, and environmental constraints. The agricultural-livestock trade system in the Indonesia-Timor Leste border region reflects an inherent trade-off between economic objectives, namely increasing income and cost efficiency, and environmental objectives, namely controlling carbon emissions. Therefore, a Multi-Objective Linear Programming (MOLP) approach was adopted to explicitly model the existing trade-offs and generate Pareto-efficient solutions, which allowed policymakers to evaluate various scenarios of alternative development while considering sustainability constraints. Incomes and carbon emissions related to food crop and cattle production were also analyzed by a MOLP model, which incorporated cost efficiency, income maximization, and objectives of environmental sustainability. The model was adapted and developed based on previous approaches recommended by Bacchetti et al. (2021), Wangsa et al. (2023), and Yuniarti et al. (2023). The finalized model aims to (1) increase production and income from rice, corn, and cattle commodities; (2) achieve total cost efficiency in trade activities; and (3) reduce carbon emissions associated with the commodity trade system.

The MOLP model is formulated as follows:

Objective Function:

$$\max Z_1 = \sum_{t=1}^T \sum_{r=1}^R \sum_{i=1}^I p_{irt} X_{irt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{j=1}^J p_{jrt} X_{jrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{k=1}^K p_{krt} X_{krt} \quad (1)$$

$$\begin{aligned} \min Z_2 = & \sum_{t=1}^T \sum_{r=1}^R \sum_{i=1}^I c_{irt} X_{irt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{j=1}^J c_{jrt} X_{jrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{k=1}^K c_{krt} X_{krt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{m=1}^M c_{mrt} Y_{mrt} \\ & + \sum_{t=1}^T \sum_{r=1}^R \sum_{n=1}^N c_{nrt} Y_{nrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{q=1}^Q c_{qrt} Y_{qrt} \end{aligned} \quad (2)$$

$$\begin{aligned} \min Z_3 = & \alpha_1 \sum_{i=1}^I X_{irt} + \alpha_2 \sum_{j=1}^J X_{jrt} + \alpha_3 \sum_{k=1}^K X_{krt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{i=1}^I \beta_t X_{irt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{j=1}^J \beta_t X_{jrt} \\ & + \sum_{t=1}^T \sum_{r=1}^R \sum_{k=1}^K \beta_t X_{krt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{m=1}^M \beta_t Y_{mrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{n=1}^N \beta_t Y_{nrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{q=1}^Q \beta_t Y_{qrt} \end{aligned} \quad (3)$$

Constraint Function:

$$\sum_{t=1}^T \sum_{r=1}^R X_{irt} \leq S_i, \forall i \quad (4)$$

$$\sum_{t=1}^T \sum_{r=1}^R X_{jrt} \leq S_j, \forall j \quad (5)$$

$$\sum_{t=1}^T \sum_{r=1}^R X_{krt} \leq S_k, \forall k \quad (6)$$

$$\sum_{t=1}^T \sum_{r=1}^R Y_{mrt} \leq S_m, \forall m \quad (7)$$

$$\sum_{t=1}^T \sum_{r=1}^R Y_{nrt} \leq S_n, \forall n \quad (8)$$

$$\sum_{t=1}^T \sum_{r=1}^R Y_{qrt} \leq S_q, \forall q \quad (9)$$

$$\sum_{t=1}^T \sum_{i=1}^I X_{irt} + \sum_{t=1}^T \sum_{m=1}^M Y_{mrt} \leq D1_r, \forall r \quad (10)$$

$$\sum_{t=1}^T \sum_{j=1}^J X_{jrt} + \sum_{t=1}^T \sum_{n=1}^N Y_{nrt} \leq D2_r, \forall r \quad (11)$$

$$\sum_{t=1}^T \sum_{i=1}^I X_{irt} + \sum_{t=1}^T \sum_{m=1}^M Y_{mrt} \leq D3_r, \forall r \quad (12)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{i=1}^I X_{irt} \leq KAP_t, \forall t \quad (13)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{j=1}^J X_{jrt} \leq KAP_t, \forall t \quad (14)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{k=1}^K X_{krt} \leq KAP_t, \forall t \quad (15)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{m=1}^M Y_{mrt} \leq KAP_t, \forall t \quad (16)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{n=1}^N Y_{nrt} \leq KAP_t, \forall t \quad (17)$$

$$\sum_{t=1}^T \sum_{r=1}^R \sum_{q=1}^Q Y_{qrt} \leq KAP_t, \forall t \quad (18)$$

$$\begin{aligned} & \alpha_1 \sum_{i=1}^I X_{irt} + \alpha_2 \sum_{j=1}^J X_{jrt} + \alpha_3 \sum_{k=1}^K X_{krt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{i=1}^I \beta_t X_{irt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{j=1}^J \beta_t X_{jrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{k=1}^K \beta_t X_{krt} \\ & + \sum_{t=1}^T \sum_{r=1}^R \sum_{m=1}^M \beta_t Y_{mrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{n=1}^N \beta_t Y_{nrt} + \sum_{t=1}^T \sum_{r=1}^R \sum_{q=1}^Q \beta_t Y_{qrt} \\ & \leq 1.61 \times \text{population} \end{aligned} \quad (19)$$

$$X_{irt}, X_{jrt}, X_{krt}, Y_{mrt}, Y_{nrt}, Y_{qrt} \geq 0 \quad (20)$$

Descriptions:

i : index of rice farmers

j : index of corn farmers

k : index of cattle farmers

m : index of rice distribution warehouses

n : index of corn distribution warehouses

q : index of cattle distribution warehouses

r : index of consumers

t : index of transportation modes

Z_1 : objective function for the income of food crop and cattle farmers

Z_2 : objective function for the total cost in the supply chain system

Z_3 : objective function for carbon emissions released into the air from supply chain activities

α_1 : carbon productivity level from rice farming

α_2 : carbon productivity level from corn farming

α_3 : carbon productivity level from cattle farming
 $z\beta$: carbon productivity level resulting from the use of transportation t
 X_{irt} : rice produced by farmer i distributed to consumer r using transportation t
 X_{jrt} : corn produced by farmer j distributed to consumer r using transportation t
 X_{krt} : cattle produced by farmer k distributed to consumer r using transportation t
 Y_{mrt} : rice in distribution warehouse m distributed to consumer r using transportation t
 Y_{nrt} : corn in distribution warehouse n distributed to consumer r using transportation t
 Y_{lrt} : cattle in distribution warehouse l distributed to consumer r using transportation t
 Y_{qrt} : cattle in distribution warehouse q distributed to consumer r using transportation t
 p_{irt} : profit obtained by farmer i if rice is distributed to consumer r using transportation t
 p_{jrt} : profit obtained by farmer j if corn is distributed to consumer r using transportation t
 p_{krt} : profit obtained by farmer k if cattle is distributed to consumer r using transportation t
 c_{irt} : transportation cost incurred to farmer i if rice is distributed to consumer r using transportation t
 c_{jrt} : transportation cost incurred to farmer j if corn is distributed to consumer r using transportation t
 c_{krt} : transportation cost incurred to farmer k if cattle is distributed to consumer r using transportation t
 c_{mrt} : transportation cost incurred to distribution warehouse m if rice is distributed to consumer r using transportation t
 c_{nrt} : transportation cost incurred to distribution warehouse n if corn is distributed to consumer r using transportation t
 c_{qrt} : transportation cost incurred by cattle distribution warehouse q when cattle are distributed to consumer r using transportation mode t
 α_1 : carbon production level from rice farming
 α_2 : carbon production level from corn farming
 α_3 : carbon production level from cattle farming
 β_t : carbon emission coefficient associated with the mode of transportation t
 S_i : production capacity of rice farming from farmer i
 S_j : production capacity of corn farming from farmer j
 S_k : production capacity of cattle farming from farmer k
 S_m : storage capacity of rice distribution warehouse - m
 S_n : storage capacity of corn distribution warehouse - n
 S_q : storage capacity of cattle distribution warehouse - q
 D_{1r} : demand for rice by consumer r
 D_{2r} : demand for corn by consumer r
 D_{3r} : demand for cattle by consumer r
 KAP_t : transport capacity of transportation mode t

The α and β parameters were calibrated using region-specific emission factors derived from Intergovernmental Panel on Climate Change (IPCC) guidelines and field survey data. Production-related emissions (α) account for the use of fertilizer, land preparation machinery, and livestock management, while transport emissions (β) are estimated based on vehicle type, carrying capacity, and distance travelled.

To analyze the performance of sustainable integrated trade between food crops (rice and corn) and cattle in the Indonesia–Timor Leste border region, the Partial Least Squares Structural Equation Modeling (PLS-SEM) approach was applied due to its predictive capability (Chin, 2009; Hair et al., 2019a; Hair et al., 2021). The conceptual framework of this study, including all latent variables and their corresponding observed indicators, is presented in Figure 2. Farmers' production is reflected by cultivated land area (Pr11), amount of seeds (Pr12), amount of fertilizer (Pr13), labor input (Pr14), and fixed costs (Pr15), while farmers' marketing is represented by marketing distance (Pe11), marketing costs (Pe12), selling price (Pe13), quantity sold (Pe14), and sales frequency (Pe15). The social dimension is indicated by motivation (S1), farmer group participation (S2), cooperation (S3), social control (S4), and the ability to access information (S5). Farmers' sustainability performance is suggested by CO₂ emissions (P1), CH₄ emissions (P2), N₂O emissions (P3), farm income (P4), and investment growth (P5). Livestock traders' production is mirrored by the number of cattle purchased (Pr21), pen capacity (Pr22), production costs (Pr23), and labor costs (Pr24), while their marketing performance is shown by transportation costs (Pe21), travel distance (Pe22), transport capacity (Pe23), and vehicle age (Pe24). Moreover, traders' sustainability performance is reflected by CO emissions (Pt1), HC levels (Pt2), CO₂ levels (Pt3), and revenue (Pt4). Food crop traders' production is exhibited by the amount of harvest purchased (Pr31), warehouse capacity (Pr32), depreciation costs (Pr33), and labor costs (Pr34), while their marketing is reflected by transportation costs (Pe31), travel distance (Pe32), transport capacity (Pe33), and vehicle age (Pe34). Apart from these, their sustainability performance is manifested by CO emissions (Pp1), HC levels (Pp2), CO₂ levels (Pp3), and revenue (Pp4). The processing industry is assessed by travel distance (IP1), transport capacity (IP2), CO emissions (IP3), and HC levels (IP4), while consumer performance is reflected by consumer satisfaction (K1), travel distance (K2), CO

emissions (K3), and HC levels (K4).

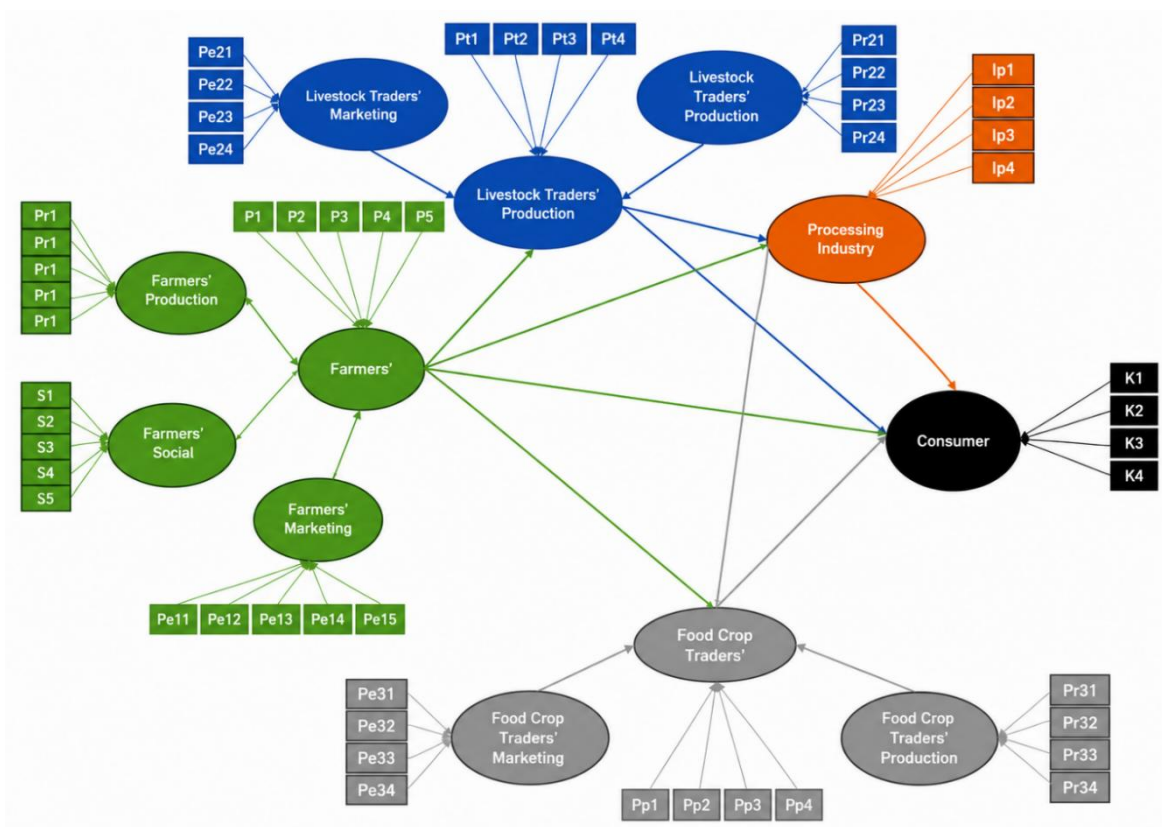


Figure 2. Hypothetical framework of sustainable trade performance of three commodities in the Indonesia–Timor Leste border region

3. Results

3.1 Optimization of Income and Carbon Emissions at the Farmer and Trader Levels

The results indicated that farmers' income could increase by approximately IDR 5,779,300,000 per planting season or livestock production cycle, equivalent to an average of around IDR 1,200,000 per month, with a production cost of IDR 64,000 per acre. Production growth can be promoted through both intensification and extensification of rice and corn cultivation. For traders, income is expected to increase by approximately IDR 8,526,000,000, with maximum carbon emissions of 2,443,241 tons CO₂e and average transport costs of IDR 4,600 per km. The optimization results for farmers' and traders' income based on considerations of carbon emissions are presented in Table 1.

Table 1. Results of income optimization and carbon emissions in the Indonesia–Timor Leste border region

No.	Actor	Earning (IDR)	Carbon Emissions (ton CO ₂ e)	Distance (km)	Cost (IDR)
1	Farmer	5,779,300,000	356.561	-	64,000
2	Trader	8,526,000,000	2443.241	767	4,600

Source: Primary data; processed in 2025.

3.2 Integrated Agricultural–Livestock Trade Performance among Three Sustainable Commodities in the Indonesia–Timor Leste Border Region

3.2.1 Validity and reliability testing

a. Convergent validity

The convergent validity test indicated that each indicator adequately represented its corresponding variable. All indicators used in the model had outer loading values greater than 0.7, which confirmed that they appropriately

reflected their latent variables. Indicators with outer loading values below 0.7 were removed from the model, as this study employed a reflective measurement model that emphasized predictive capability (Ghozali, 2021; Hair et al., 2021). The outer loading values of the indicators used in this study are presented in Table 2.

Table 2. Outer loading values of each variable and indicator

Variable	Indicator	Outer Loadings	Remarks
Processing Industry	Transport Capacity (IP1)	0.829	Valid
	Amount of CO Emissions (IP3)	0.967	Valid
	Amount of Hydrocarbon Emissions (IP4)	0.966	Valid
Consumers	Amount of CO Emissions (K3)	0.986	Valid
	Amount of Hydrocarbon Emissions (K4)	0.963	Valid
Farmers	CO ₂ Emissions (P1)	0.916	Valid
	N ₂ O Emissions (P3)	0.923	Valid
Farmers' Marketing	Quantity Sold (Pe14)	0.902	Valid
	Sales Frequency (Pe15)	0.85	Valid
Livestock Traders' Marketing	Transportation Cost (Pe21)	0.936	Valid
	Distance Travelled (Pe22)	0.927	Valid
Food Traders' Marketing	Transportation Cost (Pe31)	0.981	Valid
	Distance Travelled (Pe32)	0.981	Valid
	Transport Capacity (Pe33)	0.707	Valid
Food Traders	CO Emissions (Pp1)	0.942	Valid
	Hydrocarbon Emissions (Pp2)	0.921	Valid
	CO ₂ Concentration (Pp3)	0.993	Valid
Farmers' Production	Amount of Fertilizer (Pr13)	0.99	Valid
	Fixed Costs (Pr15)	0.711	Valid
Livestock Traders' Production	Pen Capacity (Pr22)	0.769	Valid
	Labor Costs (Pr24)	0.874	Valid
Food Traders' Production	Amount of Harvest Purchased (Pr31)	0.888	Valid
	Labor Costs (Pr34)	0.738	Valid
	CO Emissions (Pt1)	0.976	Valid
Livestock Traders	Hydrocarbon Emissions (Pt2)	0.985	Valid
	CO ₂ Concentration (Pt3)	0.956	Valid
Farmers' Social	Farmer Group Participation (S2)	0.964	Valid
	Cooperation (S3)	0.898	Valid

Source: Primary data processed, 2025.

b. Composite reliability

Composite reliability is used to measure the consistency of indicators in representing their respective variables. A variable is considered to have acceptable reliability if its composite reliability value is greater than 0.7 (Ghozali, 2021; Hair et al., 2019b; Monecke & Leisch, 2012). The composite reliability values for each variable are presented in Table 3.

Table 3. Composite reliability value

Variable	Composite Reliability	Remarks
Consumers	0.798	Reliable
Farmers	0.916	Reliable
Food Traders	0.891	Reliable
Livestock Traders	0.982	Reliable
Farmers' Marketing	0.869	Reliable
Food Traders' Marketing	0.925	Reliable
Livestock Traders' Marketing	0.783	Reliable
Processing Industry	0.945	Reliable
Farmers' Production	0.898	Reliable
Food Traders' Production	0.799	Reliable
Livestock Traders' Production	0.808	Reliable
Farmers' Social	0.929	Reliable

Source: Primary data processed, 2025.

3.2.2 Goodness-of-fit test

The goodness-of-fit of the PLS-SEM model was assessed using the R^2 , Q^2 , and f^2 values. The R^2 value indicates the extent to which the exogenous constructs explain variance in the endogenous latent variables (Ghozali, 2021; Guenther et al., 2023). The consumer variable had an R^2 value of 0.24, meaning that 24% of its variance was explained by the model. The farmer variable had an R^2 value of 0.462, indicating 46.2% explanatory power. The

food trader variable had an R^2 value of 0.962, while the livestock trader variable had an R^2 value of 0.978, suggesting that both were strongly explained by their respective construct variables. The processing industry variable showed an R^2 value of 0.22, indicating 22% explanatory capability.

Predictive relevance was evaluated using the Q^2 value. A Q^2 value greater than zero signifies that the model has predictive relevance and is therefore suitable for prediction purposes (Hair et al., 2011; Leguina, 2015). The Q^2 values for all latent variables in this study exceeded zero, confirming the predictive adequacy of the model. The R^2 and Q^2 results are presented in Table 4.

Table 4. R^2 and Q^2 value of each variable

Variable	R^2	Q^2
Consumers	0.24	0.18
Farmers	0.462	0.428
Food Traders	0.962	0.96
Livestock Traders	0.978	0.977
Processing Industry	0.22	0.17

Source: Primary data processed, 2025.

The f^2 value measures the effect size of exogenous variables on endogenous variables within the model. An f^2 value of 0.02 indicates a weak effect, 0.15 indicates a moderate effect, and 0.35 indicates a strong effect. Values below 0.02 are considered to have no meaningful influence. The f^2 values for each construct in this study are presented in Table 5.

Table 5. F^2 value of each latent variable

Latent Variable	F^2
Farmers → Consumers	0.02
Farmers → Food Traders	0.04
Farmers → Livestock Traders	0.02
Farmers → Processing Industry	0.181
Food Traders → Consumers	0.06
Livestock Traders → Consumers	0.02
Farmers' Marketing → Farmers	X'
Food Traders' Marketing → Food Traders	21.897
Livestock Traders' Marketing → Livestock Traders	33.648
Processing Industry → Consumers	0.05
Farmers' Production → Farmers	0.766
Food Traders' Production → Food Traders	0.04
Livestock Traders' Production → Livestock Traders	0.056
Food Traders → Processing Industry	0.04
Livestock Traders → Consumers	0.03
Farmers' Social → Farmers	0.206

Source: Primary data processed, 2025.

The farmer variable had a weak influence on the variables of food traders, livestock traders, and consumers, but it had a moderate influence on the processing industry variable. Among the determinants of the farmer variable, farmers' marketing showed a weak effect, farmers' production had a strong effect, and social factors had a moderate effect on farmer outcomes.

Similarly, both the livestock trader and food trader variables exerted only weak influences on the processing industry and consumer variables. For both trader types, the marketing construct showed a strong influence on the trader variable, while the production construct showed only a weak influence.

3.2.3 Sustainable integrated agricultural–livestock trade performance

The PLS-SEM analysis produced both direct and indirect effect estimates. The direct effects are presented in Table 6, while the indirect effects are presented in Table 7.

Table 6. Results of direct effects on sustainable integrated agricultural–livestock trade performance in the Indonesia–Timor Leste border region

Variable	Original Sample	t-Statistics	p-Values
Farmers → Consumers	0.04	0.441	0.659
Farmers → Food Traders	0.002	0.149	0.882
Farmers → Processing Industry	0.365	2.994	0.003 ^a
Farmers → Livestock Traders	0.011	0.804	0.422
Food Traders → Consumers	-0.122	1.177	0.239
Food Traders → Processing Industry	-0.015	0.285	0.776
Food Traders' Marketing → Consumers	-0.116	1.18	0.238
Food Traders' Marketing → Food Traders	0.949	68.126	0.000 ^a
Food Traders' Marketing → Processing Industry	-0.015	0.285	0.776
Food Traders' Production → Consumers	-0.007	0.904	0.366
Food Traders' Production → Food Traders	0.059	2.096	0.036 ^a
Food Traders' Production → Processing Industry	-0.001	0.265	0.791
Processing Industry → Consumers	0.101	0.998	0.318
Livestock Traders → Consumers	0.075	0.743	0.458
Livestock Traders → Processing Industry	-0.117	2.05	0.04 ^a
Livestock Traders' Marketing → Consumers	0.075	0.744	0.457
Livestock Traders' Marketing → Processing Industry	-0.116	2.056	0.04 ^a
Livestock Traders' Marketing → Livestock Traders	0.996	92.986	0.000 ^a
Livestock Traders' Production → Consumers	-0.002	0.63	0.529
Livestock Traders' Production → Processing Industry	0.004	1.12	0.263
Livestock Traders' Production → Livestock Traders	-0.033	1.603	0.109
Farmers' Marketing → Consumers	0.007	0.376	0.707
Farmers' Marketing → Farmers	0.168	2.167	0.03 ^a
Farmers' Marketing → Food Traders	0.000	0.13	0.897
Farmers' Marketing → Processing Industry	0.061	1.717	0.086 ^b
Farmers' Marketing → Livestock Traders	0.002	0.664	0.507
Farmers' Production → Consumers	0.023	0.43	0.667
Farmers' Production → Farmers	0.574	15.526	0.000 ^a
Farmers' Production → Food Traders	0.001	0.146	0.884
Farmers' Production → Processing Industry	0.21	3.12	0.002 ^a
Farmers' Production → Livestock Traders	0.006	0.782	0.434
Farmers' Social → Consumers	0.003	0.309	0.757
Farmers' Social → Farmers	0.086	0.918	0.359
Farmers' Social → Food Traders	0.000	0.109	0.913
Farmers' Social → Processing Industry	0.032	0.783	0.433
Farmers' Social → Livestock Traders	0.001	0.533	0.594

Source: Primary data processed, 2025. *a* and *b* are significant at $\alpha = 5\%$ and $\alpha = 10\%$, respectively.

Table 7. Results of indirect effects on sustainable integrated agricultural–livestock trade performance in the Indonesia–Timor Leste border region

Variable	Original Sample	t-Statistics	p-Values
Livestock Traders → Processing Industry → Consumers	-0.012	0.717	0.473
Farmers' Production → Farmers → Processing Industry → Consumers	0.021	0.945	0.345
Farmers' Social → Farmers → Livestock Traders → Consumers	0.000	0.359	0.719
Farmers' Social → Farmers → Livestock Traders → Processing Industry	0.000	0.454	0.650
Farmers' Social → Farmers → Food Traders → Consumers	0.000	0.078	0.938
Farmers' Production → Farmers → Consumers	0.002	0.027	0.978
Farmers' Social → Farmers → Food Traders → Processing Industry	0.000	0.028	0.978
Farmers' Production → Farmers → Food Traders	0.001	0.146	0.884
Livestock Traders' Production → Livestock Traders → Processing Industry → Consumers	0.000	0.636	0.525
Farmers' Production → Farmers → Processing Industry	0.21	3.132	0.002 ^a
Farmers' Production → Farmers → Livestock Traders	0.006	0.782	0.434
Farmers → Livestock Traders → Processing Industry → Consumers	0.000	0.417	0.676
Farmers' Marketing → Farmers → Food Traders → Processing Industry → Consumers	0.000	0.027	0.979
Food Traders' Production → Food Traders → Processing Industry → Consumers	0.000	0.202	0.840
Farmers' Marketing → Farmers → Livestock Traders → Processing Industry → Consumers	0.000	0.352	0.725

Farmers' Production → Farmers → Livestock Traders → Consumers	0.001	0.445	0.656
Food Traders' Production → Food Traders → Consumers	-0.007	0.890	0.374
Livestock Traders' Production → Livestock Traders → Consumers	-0.003	0.693	0.488
Farmers' Production → Farmers → Food Traders → Consumers	0.000	0.104	0.917
Farmers' Production → Farmers → Livestock Traders → Processing Industry	-0.001	0.592	0.554
Food Traders' Production → Food Traders → Processing Industry	-0.001	0.265	0.791
Livestock Traders' Production → Livestock Traders → Processing Industry	0.004	1.120	0.263
Farmers' Marketing → Farmers → Processing Industry → Consumers	0.006	0.789	0.43
Farmers' Production → Farmers → Food Traders → Processing Industry	0.000	0.036	0.971
Livestock Traders' Marketing → Livestock Traders → Processing Industry → Consumers	-0.012	0.718	0.472
Food Traders' Marketing → Food Traders → Processing Industry → Consumers	-0.001	0.208	0.835
Farmers → Processing Industry → Consumers	0.037	0.943	0.346
Farmers' Marketing → Farmers → Livestock Traders → Consumers	0.000	0.393	0.695
Farmers' Marketing → Farmers → Food Traders → Consumers	0.000	0.093	0.926
Farmers' Marketing → Farmers → Consumers	0.000	0.024	0.981
Farmers' Marketing → Farmers → Food Traders	0.000	0.130	0.897
Farmers' Social → Farmers → Consumers	0.000	0.022	0.983
Farmers' Marketing → Farmers → Livestock Traders → Processing Industry	0.000	0.520	0.603
Farmers' Social → Farmers → Food Traders	0.000	0.109	0.913
Farmers' Marketing → Farmers → Processing Industry	0.062	1.719	0.086
Farmers' Marketing → Farmers → Livestock Traders	0.002	0.664	0.507
Farmers' Marketing → Farmers → Food Traders → Processing Industry	0.000	0.033	0.974
Farmers' Social → Farmers → Processing Industry	0.032	0.784	0.433
Farmers' Social → Farmers → Livestock Traders	0.001	0.533	0.594
Farmers → Livestock Traders → Consumers	0.001	0.458	0.647
Farmers → Food Traders → Consumers	0.000	0.106	0.915
Farmers → Livestock Traders → Processing Industry	-0.001	0.619	0.536
Food Traders' Marketing → Food Traders → Consumers	-0.114	1.155	0.248
Livestock Traders' Marketing → Livestock Traders → Consumers	0.087	0.825	0.409
Farmers' Social → Farmers → Processing Industry → Consumers	0.003	0.520	0.603
Farmers → Food Traders → Processing Industry	0.000	0.037	0.970
Food Traders' Marketing → Food Traders → Processing Industry	-0.015	0.285	0.776
Livestock Traders' Marketing → Livestock Traders → Processing Industry	-0.116	2.056	0.04 ^a
Farmers' Production → Farmers → Food Traders → Processing Industry → Consumers	0.000	0.030	0.976
Farmers' Production → Farmers → Livestock Traders → Processing Industry → Consumers	0.000	0.398	0.691
Farmers → Food Traders → Processing Industry → Consumers	0.000	0.030	0.976
Farmers' Social → Farmers → Food Traders → Processing Industry → Consumers	0.000	0.022	0.982
Farmers' Social → Farmers → Livestock Traders → Processing Industry → Consumers	0.000	0.352	0.724
Food Traders → Processing Industry → Consumers	-0.002	0.209	0.835

Source: Primary data processed, 2025. *a* is significant at $\alpha = 5\%$.

4. Discussion

Food crop farmers in the Indonesia–Timor Leste border region cultivate an average of 41 acre of land and maintain 1–2 head of cattle. Their average monthly income from food crop farming is IDR 475,000. Income from cattle is not calculated regularly because cattle are generally treated as a form of savings to meet urgent needs such as education or social obligations. Livestock are raised under a free-range system, and production inputs are seldom recorded. On average, farming and livestock activities generate approximately 32.21 kg CO₂e emissions per farmer. Annual carbon emissions in the border region remain below the optimized emission threshold, indicating that increased agricultural activity can still take place within sustainable limits. Higher incomes may influence changes in farmers' socioeconomic practices. However, carbon-intensive practices such as land clearing through burning and limited integration between crop and livestock systems pose challenges to sustainability (Movilla-Pateiro et al., 2021).

Farmers in the border region are predominantly conventional farmers with relatively low-income levels. Farmers' income is affected by limited access to agricultural tools and technology, as well as low land productivity for rice and corn. Inappropriate use of fertilizer and limited purchasing power reduce yields (Haneloy et al., 2021; Painneon et al., 2022). Besides, improper fertilizer dosage, suboptimal land management, and waste burning contribute to increased greenhouse gas emissions (Park et al., 2023; Song et al., 2023). Farmers typically sell their harvest in several batches to avoid decline in prices during peak harvest periods and to maintain financial reserves

for unexpected expenses. However, repeated transport of small quantities contributes to additional emissions due to increased transportation frequency (Yang et al., 2022; Yang, 2023). Agricultural waste can be converted into fertilizer and complete feed, which can be used to stop the burning process and is expected to have an impact on reducing carbon emissions (Ngadisih et al., 2024). Agricultural contracts can be implemented to maintain price fluctuations at the farm level and reduce the use of repeated transportation. This kind of contracts maintain price stability and simplify farmers' work (Hoang, 2021; Miyata et al., 2009).

Farmers raise beef cattle, which are mostly free-range in forests and pastures. Cattle farming in border areas contributes significantly to carbon emissions (Pazla et al., 2022). In addition to the increasing livestock population, carbon emissions are generated from livestock manure released from feed obtained from agricultural waste, which is generally high in fiber. Changing livestock farming practices from free-range to intensive systems can reduce carbon emissions by modifying the feed used and utilizing methane for biogas production (Bayat et al., 2017; Giamouri et al., 2023).

Livestock and food traders operate with an average transport capacity of 1.5–2 tons and vehicle ages averaging 8.6 years. Traders travel an average distance of 533.35 km per purchasing cycle. Livestock traders typically have a pen capacity of 16 heads, while food traders have warehouse capacities averaging 2.7 tons. When combined, their trade activities generate approximately 425 kg CO₂e emissions and longer travel distances per purchase cycle. This shows that market demand for food and livestock products remains strong. Enhancing trade efficiency is essential to maintain trade sustainability and support continued income growth among traders (Balcom et al., 2023).

Traders in the border region procure commodities by travelling extensively across the Indonesia–Timor Leste border area. Limited availability of commodities and increased competitions encourage traders to make more frequent purchasing trips and engage more assertively in negotiations. Traders often travel long distances, stay overnight during procurement trips, and conduct vehicle repairs independently. Low awareness and limited enforcement of testing related to routine vehicle emissions contribute to higher greenhouse gas emissions among traders (Hussain et al., 2022; Porter et al., 2013).

Processing industries in the Indonesia–Timor Leste border region source limited raw materials from local food and livestock traders. Instead, inputs are primarily obtained directly from farmers or imported from other regions or islands. Raw materials are delivered either directly by farmers or collected by operators in the processing industry. Transportation activities associated with this supply chain, along with the use of firewood and other biomass fuels, are major contributors to carbon emissions in the processing sector. The reliance on locally available natural fuels reflects the broader energy conditions in rural border areas (Su et al., 2023).

Trader behavior, both in food and livestock markets, influences carbon emissions in the processing industry. Increased trading activity can lead to shortages of raw materials for processors, hence disrupting production. Such disruptions may temporarily reduce emissions but also risk lowering processing industry income due to reduced output (Nayaka & Kartika, 2018; Suryani et al., 2022). Farmers in the border region act as price takers (Lanamana, 2019; Mulyana et al., 2022). Although processing industries offer better prices, their limited capacity restricts the volume of produce they can absorb (Kanza et al., 2015).

The interactions among farmers, traders, and processing industries do not substantially affect consumer behavior. Consumers generally purchase from the nearest market or processing facility and have little direct engagement with farmers. Consequently, consumer choices have limited influence on income distribution or carbon emissions within the agricultural-livestock trade network.

5. Conclusions

Trade activity in the Indonesia–Timor Leste border region has the potential to substantially increase economic gains while maintaining carbon emissions within manageable limits. Farmer income could rise by IDR 5.7793 billion per planting season, with improved production efficiency at an average cost of IDR 64,000 per acre and a maximum carbon emissions of 356.561 tons CO₂e. Trader income could increase by IDR 8.526 billion, with a maximum carbon emissions of 2,443.241 tons CO₂e and average transportation costs of IDR 4,600 per kilometer.

At the farm level, carbon emissions primarily stem from inefficient use of fertilizer, land management, and waste burning. For traders, emissions are driven by transport distance and limited vehicle capacity. Although processing industries have direct relationships with some farmers, most farmers remain dependent on traders for access to the market. Strengthening and expanding the capacity of the processing industry in the border region is therefore essential to increasing farmers' income and improving the sustainability of agricultural-livestock trade.

Despite its contributions, this study has several limitations. The analysis was limited to four districts in the Indonesia–Timor Leste border region and three commodities, i.e., rice, corn, and livestock, which may limit the generalizability of the findings. Communities in the Indonesia–Timor Leste border region are predominantly traditional farmers dependent on rainfall. Livestock farming is practiced in open pasture. Furthermore, the MOLP framework relies on linear assumptions, and carbon emission estimates do not fully capture indirect or life-cycle emissions. PLS-SEM results also depend on selected reflective indicators and self-reported survey data. Future research should expand the geographic and commodity coverage, incorporate additional supply chain actors, and

adopt dynamic or non-linear approaches to better capture the complexities of sustainable border trade systems.

Author Contributions

Conceptualization, B.P.S., U.J., F.M.A.B., and A.S.M.; methodology, B.P.S., A.N., and Y.P.V.M.; software, B.P.S., F.M.A.B., and R.A.D.; validation, B.P.S., P.D.K.P., A.S.M., and U.J.; formal analysis, B.P.S., R.A.D., F.M.A.B., and A.S.M.; investigation, B.P.S., U.J., A.S.M., F.M.A.B., and R.A.D.; resources, B.P.S., U.J., A.S.M., F.M.A.B., and R.A.D.; data curation, B.P.S., U.J., A.S.M., F.M.A.B., and R.A.D.; writing—review and editing, B.P.S., U.J., A.S.M., A.N., and P.D.K.P.; visualization, R.A.D.; supervision, B.P.S., U.J., A.S.M., F.M.A.B., Y.P.V.M., and P.D.K.P.; project administration, R.A.D.; funding acquisition, B.P.S. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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